THE ONE-COORDINATE STEP IN THE FOURIER ENTROPY-INFLUENCE CONJECTURE

YITZCHAK SHMALO

ABSTRACT. For a Boolean function f and a coordinate i let $D_i(f) = H(f) - \frac{1}{2}H(f_{i\to 1}) - \frac{1}{2}H(f_{i\to -1})$ be the drop in Fourier entropy when x_i is fixed. González, MacManus and Pereyra observed that the Fourier entropy-influence conjecture follows, by induction on the dimension, if every Boolean function has a coordinate with $D_i(f) \le K\,\mathrm{Inf}_i(f)$. We show that this condition is also necessary: if the conjecture holds with constant C, then every nonconstant Boolean function has a coordinate whose erasure entropy is at most $C\,\mathrm{Inf}_i(f)$, and hence one with $D_i(f) \le (C + 2/\ln 2)\mathrm{Inf}_i(f)$. The constant $2/\ln 2$ is the sharp bound for the Jensen–Shannon part of D_i. We also give an equivalent pseudo-likelihood form of the conjecture, a bound on the influences of extremal functions, and an example showing that the condition cannot be required at every coordinate.

1. INTRODUCTION

Let $f : \{-1,1\}^n \to \{-1,1\}$ have Fourier expansion $f = \sum_S \hat{f}(S)\chi_S$. By Parseval, $\mu(S) = \hat{f}(S)^2$ is a probability distribution on subsets of $[n]$, the spectral distribution of f. Write

$$H(f) = \sum_S \mu(S)\log_2\frac{1}{\mu(S)}, \qquad \mathrm{Inf}_i(f) = \sum_{S\ni i}\mu(S), \qquad I(f) = \sum_i \mathrm{Inf}_i(f) = \sum_S |S|\mu(S).$$

The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [2] asks for a universal C with $H(f) \le C\,I(f)$ for every Boolean f. It is known for many classes, among them symmetric functions and read-once decision trees [7], read-once formulas [6, 1], functions of bounded average decision-tree depth [10], and plateaued functions [8]; the best general bound is due to Han [4], and the min-entropy version is studied in [5].

Fix a coordinate i and the restrictions $f^{\pm} = f_{i\to\pm 1}$. Since $I(f) = \frac{1}{2}I(f^+) + \frac{1}{2}I(f^-) + \mathrm{Inf}_i(f)$, the inequality

$$D_i(f) := H(f) - \tfrac{1}{2}H(f^+) - \tfrac{1}{2}H(f^-) \le K\,\mathrm{Inf}_i(f) \tag{1}$$

for one coordinate i of every Boolean f gives $H(f) \le K\,I(f)$ by induction on n. This route is suggested by Han [4, (3.17) and the discussion after Remark 3.2] in a different local form, and is set out by González, MacManus and Pereyra [3], who prove (1) along suitable decision trees for several new classes. The inequality version $D_1(f) \le 2$ is Lemma 3.2 of Chakraborty, Kulkarni, Lokam and Saurabh [1].

Our main observation is that the route loses nothing: the one-coordinate condition is equivalent to the conjecture. Let $C_n = \max H(f)/I(f)$ over nonconstant Boolean f on n variables, and

$$R_n = \max_f \min_{i:\,\mathrm{Inf}_i(f)>0} \frac{E_i(f)}{\mathrm{Inf}_i(f)}, \qquad R'_n = \max_f \min_{i:\,\mathrm{Inf}_i(f)>0} \frac{D_i(f)}{\mathrm{Inf}_i(f)},$$

with the maxima over the same functions and E_i the erasure entropy defined in Section 2.

Theorem 1. *For every n,*

$$R_n \le C_n \le \max_{m\le n} R_m + \frac{2}{\ln 2}, \qquad \max_{m\le n} R'_m \ge C_n, \qquad R'_n \le C_n + \frac{2}{\ln 2}.$$

In particular the FEI conjecture holds if and only if $\sup_n R_n < \infty$*, if and only if* $\sup_n R'_n < \infty$.

The proof is short. The entropy drop splits into an erasure part and a Jensen–Shannon part (this is the computation in [3, Appendix B]); the Jensen–Shannon part is at most $(2/\ln 2)\mathrm{Inf}_i$, with this constant sharp (Lemma 3); and the erasure parts of all coordinates add up to at most $H(f)$ (Lemma 4). Section 4 gives the pseudo-likelihood form, the bound on extremal influences, and the example.

2. The split

Let U be a random set with law μ and identify U with its indicator vector $(X_1, \dots, X_n)$. For $T \subseteq [n]\setminus\{i\}$ write $a_T = \hat{f}(T)$ and $b_T = \hat{f}(T\cup\{i\})$. Then $\widehat{f^{\pm}}(T) = a_T \pm b_T$, so $\mu^{\pm}(T) = (a_T \pm b_T)^2$ are the spectral distributions of the two restrictions, and $\frac{1}{2}(\mu^+(T) + \mu^-(T)) = a_T^2 + b_T^2 = \Pr[U \setminus \{i\} = T]$. Let $h(q) = q\log_2 \frac{1}{q} + (1-q)\log_2 \frac{1}{1-q}$. Define

$$E_i(f) = H(X_i \mid U \setminus \{i\}) = \sum_T (a_T^2 + b_T^2)\, h\Big(\frac{b_T^2}{a_T^2 + b_T^2}\Big), \qquad J_i(f) = \mathrm{JS}(\mu^+, \mu^-),$$

where $\mathrm{JS}(p,q) = H(\frac{p+q}{2}) - \frac{1}{2}H(p) - \frac{1}{2}H(q)$ is the Jensen–Shannon divergence in bits.

Lemma 2 ([3], Appendix B)**.** $D_i(f) = E_i(f) + J_i(f)$.

Proof. By the chain rule $H(f) = H(U) = H(U \setminus \{i\}) + H(X_i \mid U \setminus \{i\})$. The law of $U \setminus \{i\}$ is $\frac{1}{2}(\mu^+ + \mu^-)$, so $H(U \setminus \{i\}) = \frac{1}{2}H(f^+) + \frac{1}{2}H(f^-) + \mathrm{JS}(\mu^+, \mu^-)$. □

Both parts are nonnegative, and $E_i(f) \le H(X_i) = h(\mathrm{Inf}_i(f))$.

Lemma 3. $J_i(f) \le \frac{2}{\ln 2}\mathrm{Inf}_i(f)$ *for every Boolean* f *and every coordinate* i*, and the constant is sharp.*

Proof. Write $|a_T| = r\cos\theta$, $|b_T| = r\sin\theta$ with $\theta \in [0, \pi/2]$, and $s = \sin 2\theta$. The term of T in J_i is $r^2\big(1 - h(\frac{1+\sigma s}{2})\big)$ with $\sigma = \pm 1$ the sign of $a_T b_T$; since h is symmetric, it equals $r^2\varphi(s)/\ln 2$, where

$$\varphi(s) = \tfrac{1}{2}\big[(1+s)\ln(1+s) + (1-s)\ln(1-s)\big].$$

The term of T in Inf_i is $b_T^2 = r^2\sin^2\theta$, so it suffices to show $\varphi(\sin 2\theta) \le 2\sin^2\theta$. For $\theta \ge \pi/4$ this holds because $\varphi \le \ln 2 < 1 \le 2\sin^2\theta$. On $[0, \pi/4]$ put $F(\theta) = 2\sin^2\theta - \varphi(\sin 2\theta)$, which is continuous. Then $F(0) = 0$ and, on $[0, \pi/4)$, since $\varphi'(s) = \operatorname{artanh} s$ and $\cos 2\theta = \sqrt{1-s^2}$ there,

$$F'(\theta) = 2s - 2\sqrt{1-s^2}\,\operatorname{artanh}(s).$$

With $s = \tanh y$, $y \ge 0$, this is $2\tanh y - 2y/\cosh y = 2(\sinh y - y)/\cosh y \ge 0$. Hence $F \ge 0$ on $[0, \pi/4]$.

For sharpness, $\varphi(s) = s^2/2 + O(s^4)$, so $\varphi(\sin 2\theta)/(2\sin^2\theta) \to 1$ as $\theta \to 0$. Take a Boolean f and coordinate i with $b_T^2/(a_T^2 + b_T^2) \to 0$ uniformly in T along a sequence, for example the function of Proposition 7 below; then $J_i/\mathrm{Inf}_i \to 2/\ln 2$. □

The bound in [3, Lemma 8.1] is $1 - h(t) \le 4(\frac{1}{2} - t)^2$, which gives $J_i \le 4\,\mathrm{Inf}_i$.

Lemma 4. $\sum_{i=1}^n E_i(f) \le H(f)$.

Proof. $H(f) = H(X_1, \dots, X_n) = \sum_i H(X_i \mid X_1, \dots, X_{i-1}) \ge \sum_i H(X_i \mid X_j,\, j \ne i) = \sum_i E_i(f)$, since conditioning on more variables does not increase entropy. □

3. Proof of Theorem 1

$R_n \le C_n$. Let f be nonconstant on n variables. By Lemma 4, $\sum_i E_i(f) \le H(f) \le C_n I(f) = C_n \sum_i \mathrm{Inf}_i(f)$. The coordinates with $\mathrm{Inf}_i(f) = 0$ have $X_i = 0$ almost surely and $E_i(f) = 0$, so some coordinate with $\mathrm{Inf}_i(f) > 0$ has $E_i(f) \le C_n \mathrm{Inf}_i(f)$.

$C_n \le \max_{m \le n} R_m + 2/\ln 2$. Let $K = \max_{m \le n} R_m + 2/\ln 2$. We show $H(g) \le K\, I(g)$ for every Boolean g on $m \le n$ variables by induction on the number of coordinates g depends on. If it depends on none, $H(g) = I(g) = 0$. Otherwise choose i with $\mathrm{Inf}_i(g) > 0$ and $E_i(g) \le R_m \mathrm{Inf}_i(g)$. By Lemmas 2 and 3, $D_i(g) \le K \mathrm{Inf}_i(g)$; the restrictions $g^{\pm}$ depend on fewer coordinates, so

$$H(g) = \tfrac{1}{2}H(g^+) + \tfrac{1}{2}H(g^-) + D_i(g) \le K\big(\tfrac{1}{2}I(g^+) + \tfrac{1}{2}I(g^-) + \mathrm{Inf}_i(g)\big) = K\, I(g).$$

$\max_{m \le n} R'_m \ge C_n$ is the same induction with D_i in place of E_i and no Jensen–Shannon term, which is the argument of [3]. $R'_n \le C_n + 2/\ln 2$: by Lemmas 2, 3 and 4, $\sum_i D_i(f) \le H(f) + \frac{2}{\ln 2} I(f) \le (C_n + \frac{2}{\ln 2}) I(f)$, and a coordinate with $\mathrm{Inf}_i(f) > 0$ attains the average. □

4. Further consequences

A pseudo-likelihood form. The quantity $\sum_i E_i(f) = \sum_i H(X_i \mid X_j,\, j \ne i)$ is the pseudo-likelihood entropy of the spectral sample: the expected log-loss, in bits, of predicting each coordinate of U from all the others.

Corollary 5. *If $H(f) \le C\, I(f)$ for all Boolean f, then $\sum_i E_i(f) \le C\, I(f)$ for all Boolean f. Conversely, if $\sum_i E_i(f) \le C\, I(f)$ for all Boolean f, then $H(f) \le (C + 2/\ln 2)\, I(f)$ for all Boolean f.*

Proof. The first statement is Lemma 4. For the second, the hypothesis gives $R_n \le C$ for all n by averaging, and Theorem 1 applies. □

So FEI may be read as a statement about conditional rather than joint uncertainty of the spectral sample: it is equivalent to asking that each element of U be predictable from the rest at a cost of $O(1)$ bits per expected element.

Extremal functions have small influences.

Proposition 6. *Let f be a Boolean function on n variables with $H(f) = C_n I(f)$. Then every coordinate with $\mathrm{Inf}_i(f) > 0$ satisfies*

$$\mathrm{Inf}_i(f) \le 2^{\,3/\ln 2 - C_{n-1}}.$$

Proof. Let $p = \mathrm{Inf}_i(f) > 0$. The restrictions are functions of $n-1$ variables, so $H(f^{\pm}) \le C_{n-1} I(f^{\pm})$ (trivially if they are constant), and $C_{n-1} \le C_n$, so

$$D_i(f) = C_n I(f) - \tfrac{1}{2}H(f^+) - \tfrac{1}{2}H(f^-) \ge C_n I(f) - C_{n-1}\big(I(f) - p\big) \ge C_{n-1}\, p.$$

On the other side $D_i(f) \le h(p) + \frac{2}{\ln 2} p$, and $h(p) \le p \log_2 \frac{1}{p} + \frac{p}{\ln 2}$ because $(1-p)\ln\frac{1}{1-p} \le p$. Dividing by p gives $C_{n-1} \le \log_2 \frac{1}{p} + \frac{3}{\ln 2}$. □

Thus if $C_n \to \infty$, every coordinate of every extremal function has influence tending to zero. This is not true of arbitrary functions with a large ratio H/I: the function $f(x)\, x_{n+1}$ has the same entropy as f, total influence $I(f) + 1$, and a coordinate with $D_{n+1} = 0$.

The condition cannot be asked of every coordinate.

Proposition 7. *Let k be even, g a bent function on $\{-1,1\}^k$, y^* a point, and define f on $\{-1,1\}\times\{-1,1\}^k$ by $f(x_0,y)=g(y)$ for $y\neq y^*$ and $f(x_0,y^*)=x_0$. Then $\mathrm{Inf}_0(f)=2^{-k}$ and*

$$\frac{E_0(f)}{\mathrm{Inf}_0(f)} = k + \log_2 e + O\Big(2^{-k/2}\Big).$$

Proof. Since $f = g + \mathbf{1}_{y^*}(y)\,(x_0 - g(y^*))$ and $\mathbf{1}_{y^*} = 2^{-k}\sum_T \chi_T(y^*)\chi_T$, for $T\subseteq[k]$ we get $b_T = 2^{-k}\chi_T(y^*)$ and $a_T = \hat g(T) - 2^{-k}g(y^*)\chi_T(y^*)$. Because g is bent, $\hat g(T)^2 = 2^{-k}$, so $b_T^2 = 4^{-k}$ and $a_T^2 = 2^{-k}(1\pm 2^{-k/2})^2$. Hence $\mathrm{Inf}_0(f) = \sum_T b_T^2 = 2^{-k}$, and every $q_T = b_T^2/(a_T^2+b_T^2)$ equals $2^{-k}(1+O(2^{-k/2}))$, so $\log_2(1/q_T) = k + O(2^{-k/2})$. Using $h(q) = q\log_2\frac{1}{q} + q\log_2 e + O(q^2)$,

$$E_0(f) = \sum_T (a_T^2+b_T^2)\,h(q_T) = \sum_T b_T^2\Big(\log_2\frac{1}{q_T} + \log_2 e + O(q_T)\Big) = 2^{-k}\big(k + \log_2 e + O(2^{-k/2})\big). \ \square$$

The ratio is unbounded while $I(f)\approx k/2$ and $H(f)\approx k$, so FEI is not in danger: the other coordinates are good. For $k=2,4,6,8,10$ the exact values of E_0/Inf_0 are 2.672, 5.224, 7.387, 9.429 and 11.439. The same function shows that the ratio J_0/Inf_0 tends to $2/\ln 2$, which completes the proof of Lemma 3.

Remark 8. A sufficient condition used in [3, Corollary 4.3] is $\sum_T |\mu^+(T)-\mu^-(T)| \le K\mathrm{Inf}_i(f)$ for one coordinate, which is Han's Question 3.1 [4], since $|\mu^+(T)-\mu^-(T)| = 4|a_T b_T|$. This fails at every coordinate of some read-once formulas, with ratio at least the depth [9]; there the one-coordinate entropy step (1) still holds with $K = (5+\ln 4)/\ln 2$. The erasure condition behaves differently: by Lemma 4, every nonconstant Boolean f with $H(f)\le C\,I(f)$ has a coordinate with $E_i(f)\le C\,\mathrm{Inf}_i(f)$, so it cannot fail on any class where FEI holds.

5. Computations

The identity of Lemma 2, the bound of Lemma 3 and the values in Proposition 7 were checked numerically: the identity on all Boolean functions of four variables and on 40,000 random functions each of five, six and seven variables at three densities, with D_i computed from the restricted truth tables and E_i, J_i from the Fourier coefficients (largest discrepancy $7\cdot 10^{-15}$); the largest observed J_i/Inf_i was $2.806 < 2/\ln 2 \approx 2.885$. On all functions of four variables $\max_f \min_i E_i/\mathrm{Inf}_i = 2.672$, and on the random sample of five variables it was 2.696. The scripts are in the repository accompanying this note.

Acknowledgements and funding

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [11] and working together inside the Brain framework [13], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [12], the Math Brain repository [11] and the Brain framework repository [13].

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [13] and Math Brain [11]. In November 2026 a Claude Fable agent (Anthropic), connected to the Brain and working alongside Codex agents running OpenAI's GPT-6.1 on the same problem, derived the one-coordinate form of the conjecture, the averaging argument over coordinates and the bent point-replacement example, and wrote the first numerical checks. Claude Opus 5.5 (Anthropic) searched the literature, proved the sharp constant in Lemma 3, wrote independent code for every numerical claim, prepared the manuscript and audited it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

`https://github.com/yspennstate/fourier-entropy-influence-one-coordinate`

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [11].

References

[1] S. Chakraborty, R. Kulkarni, S. V. Lokam, N. Saurabh, Upper bounds on Fourier entropy, Electronic Colloquium on Computational Complexity TR13-052, 2013.

[2] E. Friedgut, G. Kalai, Every monotone graph property has a sharp threshold, Proc. Amer. Math. Soc. 124 (1996), 2993–3002.

[3] M. J. González, P. MacManus, M. C. Pereyra, Further evidence towards the Fourier entropy-influence conjecture, arXiv:2606.00246, 2026.

[4] X. Han, A new bound for the Fourier-entropy-influence conjecture, Combinatorica 45 (2025), arXiv:2312.08271.

[5] E. Kelman, G. Kindler, N. Lifshitz, D. Minzer, M. Safra, Towards a proof of the Fourier-entropy conjecture?, Geom. Funct. Anal. 30 (2020), 1097–1138.

[6] R. O'Donnell, L.-Y. Tan, A composition theorem for the Fourier entropy-influence conjecture, ICALP 2013, arXiv:1304.1347.

[7] R. O'Donnell, J. Wright, Y. Zhou, The Fourier entropy-influence conjecture for certain classes of Boolean functions, ICALP 2011.

[8] V. N. Potapov, On the Fourier entropy-influence conjecture for Boolean plateaued functions, arXiv:2609.21563, 2026.

[9] Y. Shmalo, Read-once formulas and a question of Han on the Fourier entropy-influence conjecture, `https://github.com/yspennstate/han-question-read-once-formulas`, 2026.

[10] A. Wan, J. Wright, C. Wu, Decision trees, protocols, and the Fourier entropy-influence conjecture, ITCS 2014, arXiv:1312.3003.

[11] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[12] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[13] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.